

FROM DATA DISCOVERY TO FEDERATED ANALYSIS

Supporting the Research Journey

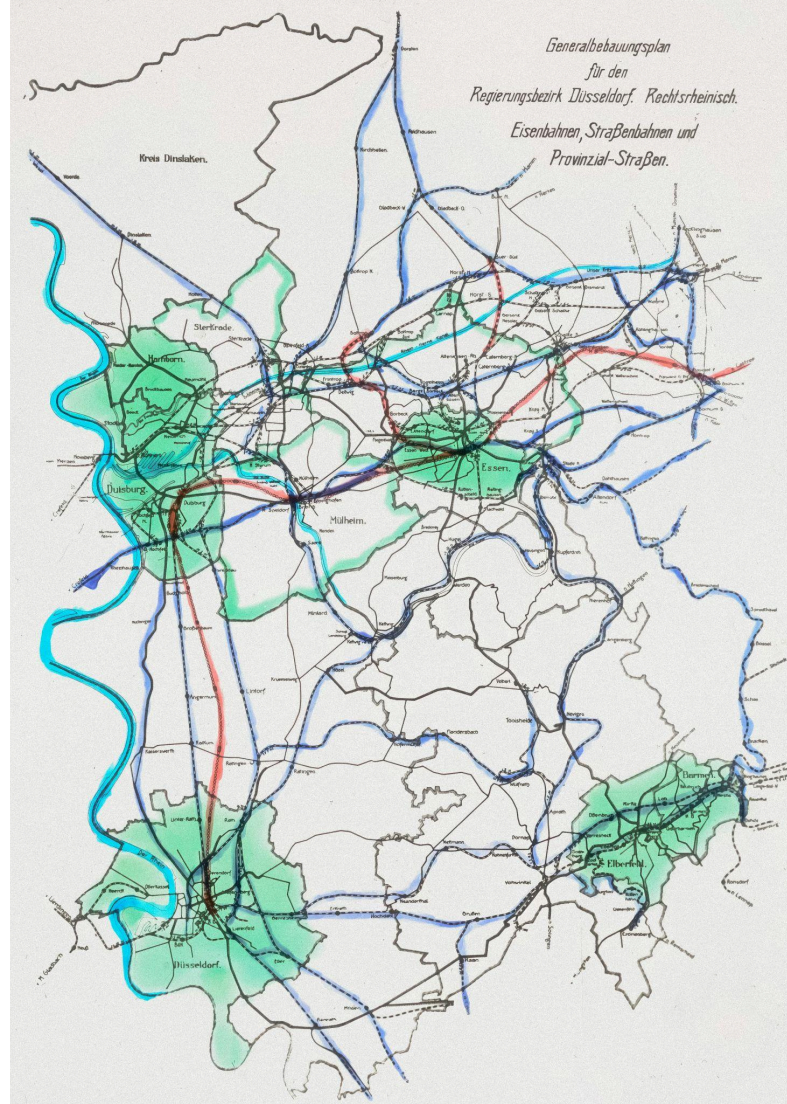
Tim Cadman

Senior Data Scientist

THE PROBLEM

A fragmented researcher journey

- Large quantities of existing data (e.g. cohort studies, biobanks)
- Combining data sources increases statistical power and enables replication
- However, researchers face a fragmented landscape when trying to find and reuse data



OUR JOURNEY

From FAIR at Source to FAIR Research Federations

	Challenge	Solution
Local data	Not structured or reusable	Harmonise to established open standards
Catalogue	Collections not discoverable	Open source metadata catalogue
Request	Months lost negotiating collection-by-collection	Central point of access for all requests
Access	Each holder manages permissions separately	Federated identity & access
Analyse	Data transfer carries ethical, legal and practical difficulties	Federated analysis — data never moves

OUR APPROACH

One open stack – same at every scale

Local data

Harmonise to established standards

Catalogue

MOLGENIS meta-data catalogue

Request

BBMRI-ERIC Negotiator

Access

LS AAI + OIDC

Analyse

DataSHIELD + Molgenis Armadillo

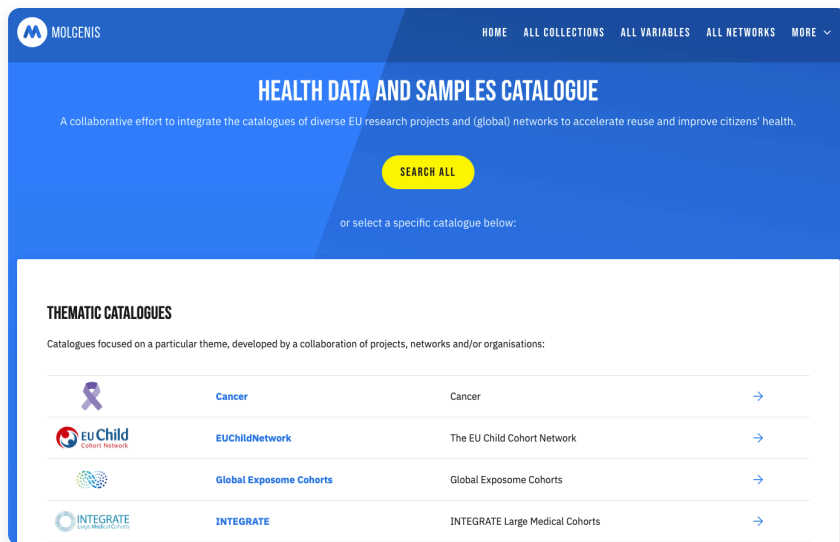
LOCAL DATA

Harmonise to established standards

- *Placeholder — content to be added*

CATALOGUE

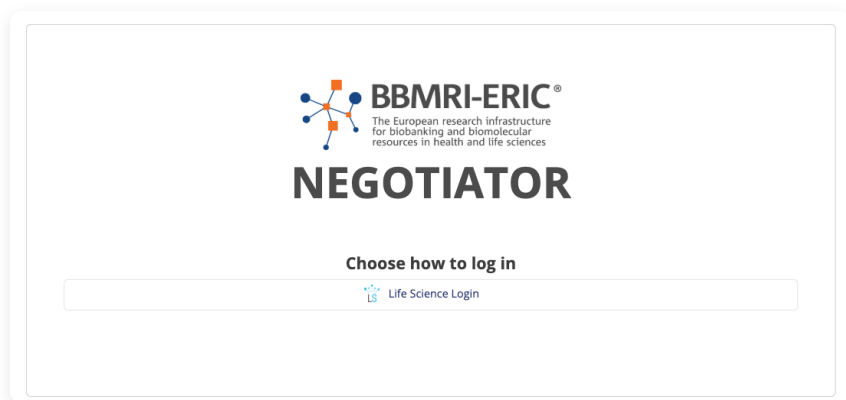
Making collections discoverable



- Open source metadata catalogue — information about data, not the data itself
- Search across consortia for harmonised variables and collections
- Upload variables and Common Data Model mappings

NEGOTIATOR

One request – not twenty emails



- Open-source access negotiation for research infrastructures
- One request fans out to many holders — each keeps its own yes/no
- Customisable workflows, messaging, and moderation

ACCESS

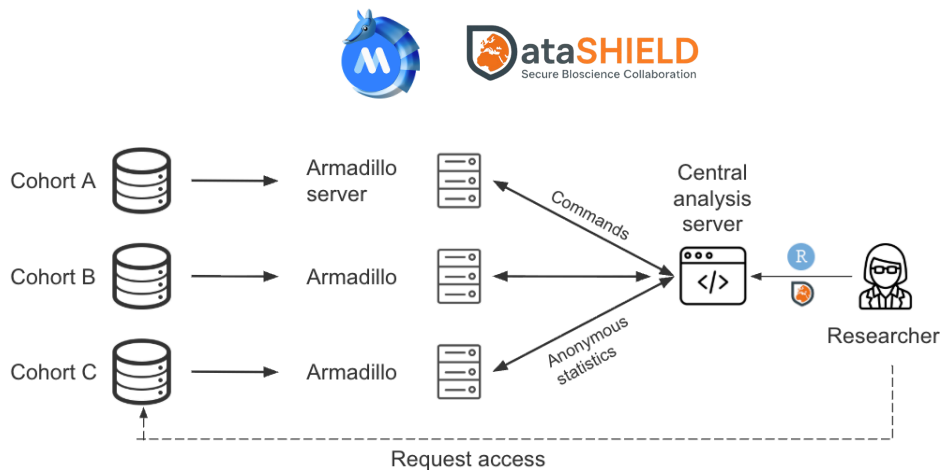
Federated identity & access



- LS Login — federated identity service from EOSC-Life
- Researchers authenticate with home institution credentials
- Single sign-on across research infrastructures — no separate accounts

ANALYSE

Federated analysis – data never moves



- Analysis travels to the data — only safe summary statistics shared
- MOLGENIS Armadillo — lightweight server per cohort
- 20+ nodes in production across LifeCycle, ATHLETE, LongITools, EUCAN-Connect

DEMO

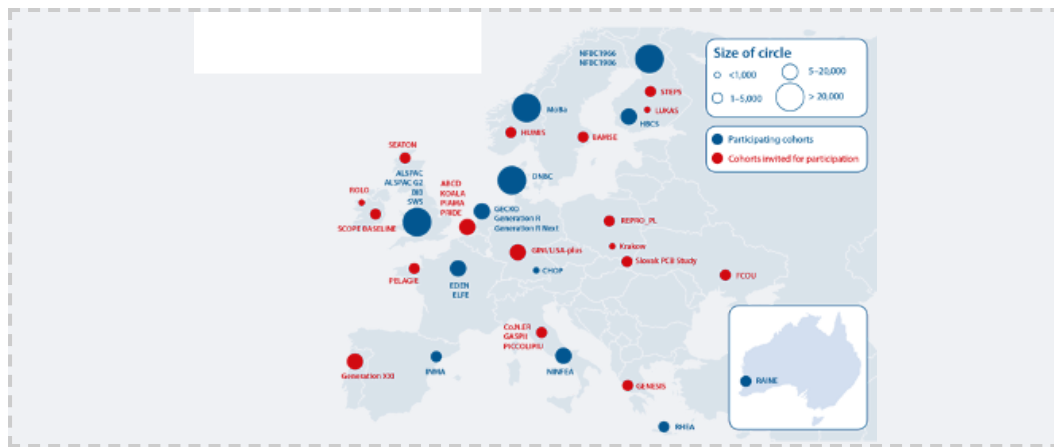
Research question

*"How is mode of delivery (e.g. vaginal, caesarean)
associated with maternal mental health across different
EU countries?"*

DEMO

Local data

Health data harmonised in previous EU 'LifeCycle' project



Local data

DEMO

Catalogue

Search for birth cohorts with relevant harmonised data

The screenshot displays the 'VARIABLES' section of the MolGenis interface. On the left, a sidebar contains a search bar with 'mode_delivery' and a 'SEARCH' button. Below the search bar are sections for 'Topics' and 'Sources', each with a list of checkboxes and icons. The main content area is titled 'VARIABLES' and 'A complete overview of harmonised variables'. It features a 'LIST OF VARIABLES' button and a 'HARMONISATIONS' button. Below these, a table lists variables with columns for 'mode_delivery', 'ABCD', 'ALSPAC', 'ATHLETE', 'BIB', 'BISC', 'CELSPAC-TNG', 'CHOP', 'CONSTANCES', 'DFBC', 'DNBC', 'EDEN', 'ELFE', 'ELPA', 'ELSPAC', 'ENVIRONAGE', 'FEF', 'FinGedi', 'G21', 'GECKO', 'GENESIS', 'GenR', 'GenR Next', 'HBCS', 'HELIX', and 'HGS'. The 'mode_delivery' column shows a blue checkmark for 'mode_delivery' and a blue checkmark for 'mode_delivery'.

mode_delivery	ABCD	ALSPAC	ATHLETE	BIB	BISC	CELSPAC-TNG	CHOP	CONSTANCES	DFBC	DNBC	EDEN	ELFE	ELPA	ELSPAC	ENVIRONAGE	FEF	FinGedi	G21	GECKO	GENESIS	GenR	GenR Next	HBCS	HELIX	HGS
mode_delivery	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Local data

Catalogue

Request

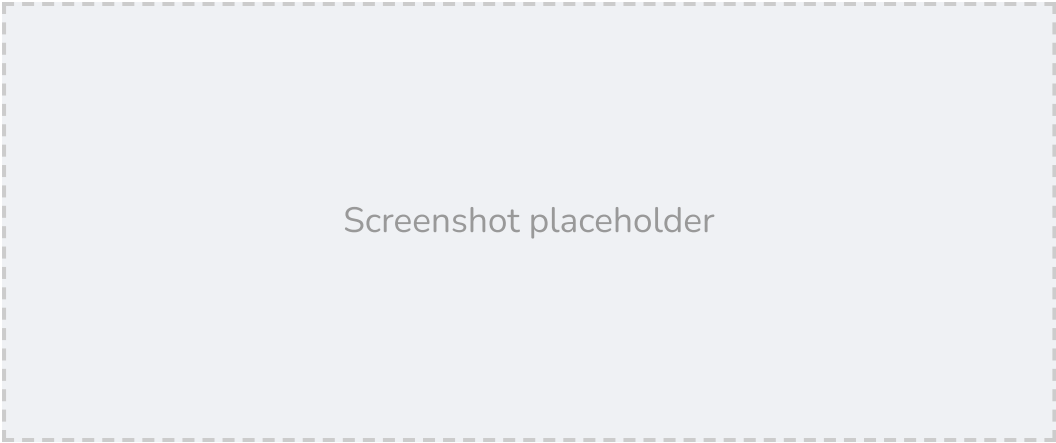
Access

Analyse

DEMO

Catalogue

TODO: feature request to select multiple variables and view in harmonisation matrix



Screenshot placeholder

Local data

Catalogue

Request

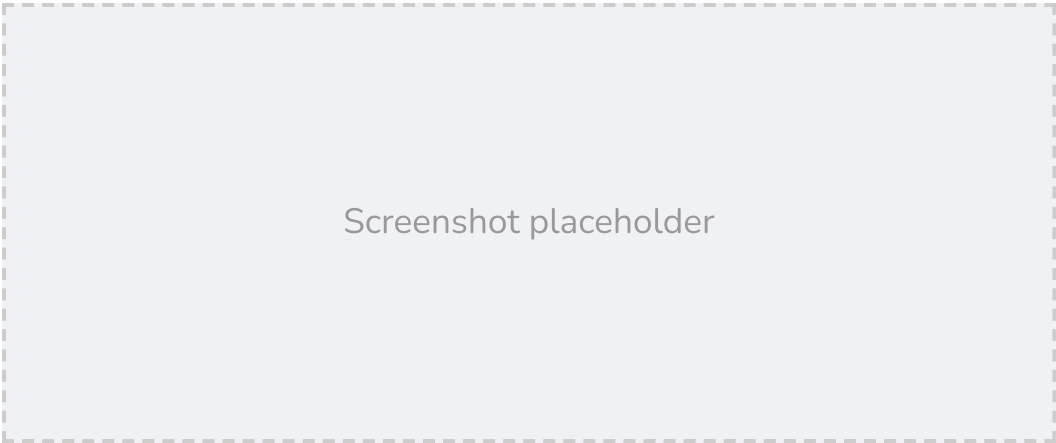
Access

Analyse

DEMO

Catalogue

TODO: feature request to send multiple variable request to Negotiator without going via BBMRI directory



Screenshot placeholder

Local data

Catalogue

Request

Access

Analyse

DEMO

Request

Enter project details

 Your negotiations Guide

Sections

-  Resources
-  **Project**
Provide information about your project
-  **Request**
Provide information the resources you are requesting
-  **Ethics vote**
Is ethics vote present in your project?
-  **Review**

Project

Provide information about your project

Title * 

Description * 

Scientific board vote * 
☒ given ☐ notgiven
Please select one of available values

Local data

Catalogue

Request

Access

Analyse

DEMO

Request

Provide study description

The screenshot shows the 'Request' form in the BBMRI-ERIC NEGOTIATOR. The top navigation bar includes the logo, 'Your negotiations', and a 'Guide' link. On the left, a 'Sections' sidebar lists: Resources (info icon), Project (checkmark icon), Request (selected, with an error icon), Ethics vote (error icon), and Review (checkmark icon). The main 'Request' section has the heading 'Request' and the instruction 'Provide information the resources you are requesting'. It features a 'Description' field with a red border and a red error message: 'Please provide a text'. The text inside the field reads: 'Background: The correlation between mode of birth and maternal psychological health has been studied, but not extensively. A systematic review (Carter et al., 2022) found a'. Below the field are three buttons: 'Save Draft', '< Back', and 'Next >'.

Local data

Catalogue

Request

Access

Analyse

DEMO

Request

Enter ethics details

 Your negotiations  Guide

Sections

-  Resources
-  Project
Provide information about your project
-  Request
Provide information the resources you are requesting
-  **Ethics vote**
Is ethics vote present in your project?
-  Review

Ethics vote

Is ethics vote present in your project?

Ethics vote * 

Please provide a text

Attachment 

Seleccionar archivo

Ningún archivo seleccionado

Save Draft

< Back

Next >

Local data

Catalogue

Request

Access

Analyse

DEMO

Request

Submit request

BBMRI-ERIC
NEGOTIATOR

Your negotiations

Guide

Sections

Resources

Project

Provide information about your project

Request

Provide information the resources you are requesting

Ethics vote

Is ethics vote present in your project?

Review

Overview

Upon confirmation, your request will undergo content review. Our reviewers may contact you via email for further details. Upon approval, the respective organizations you wish to contact will be notified of your request. Please click 'Submit request' and then 'Confirm' to proceed.

PROJECT

Title: LMedC Integrate Demonstrator

Description: How is mode of birth associated with maternal mental health?

Scientific board vote: given

REQUEST

Description: Background: The correlation between mode of birth and maternal psychological health has been studied, but not extensively. A systematic review (Carter et al., 2022) found a significant relationship between mode of birth and maternal PTSD symptoms, with caesarean birth being more closely associated with PTSD than vaginal delivery. A direct meta-analysis performed by (Sun et al., 2021) revealed that the mode of delivery has a significant effect on the occurrence of mild postnatal depression. Another meta-analysis conducted by (Nina et al., 2020)

Save Draft

< Back

Next >

Submit Request

Local data

Catalogue

Request

Access

Analyse

 MOLGENIS

DEMO

Request

Await approval

[Go back](#)

LMEDC INTEGRATE DEMONSTRATOR
UNDER REVIEW

PROJECT
title
LMedC Integrate Demonstrator
description
How is mode of birth associated with maternal mental health?
Scientific board vote
given

REQUEST
description
Background: The correlation between mode of birth and maternal psychological health has been studied, but not extensively. A systematic review (Carter et al., 2022) found a significant relationship between mode of birth and maternal PTSD symptoms, with caesarean birth being more closely associated with PTSD than vaginal delivery. A direct meta-analysis performed by (Sun et al., 2021) revealed that the mode of delivery has a significant effect on the occurrence of mild postpartum depression. Another meta-analysis conducted by (Ning et al., 2024) suggested that PTSD risk may rise in the presence of Caesarean sections, with the risk rising by 20% in the emergency CS group. (Heyne et al., 2022) investigated the prevalence of birth-related PTSD among parents (both maternal and paternal). Mean prevalence of birth-related PTSD was 4.7% in mothers and 1.2% in fathers. A study conducted on Gen R, one of the cohort studies included in our analysis, found that several pregnancy and delivery complications present a risk for women's mental health in the postpartum period (Blom et al., 2010). Despite these findings, limited research has examined how different trajectories of labor onset and mode of birth interact to influence maternal mental health outcomes.
The INTEGRATE-IMedC consortium is funded by the European Research Agency to develop a new concept to guide and support decision-making for the next generation research infrastructure (RI) to facilitate efficient utilization and harmonisation of large medical cohorts (LMedC). One of the objectives of this project is to showcase how harmonised data and federated analysis can accelerate scientific and medical breakthroughs in Europe and beyond.
Our goal for this study is to use the rich data harmonised within the EU Child Cohort Network to examine how

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Negotiation ID:
613
UUID:
2a5dc02-d1f4-4eb2-83f9-373983e5c531
Submitted at:
05.06.2025 08:43
Status:
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Local data

Catalogue

Request

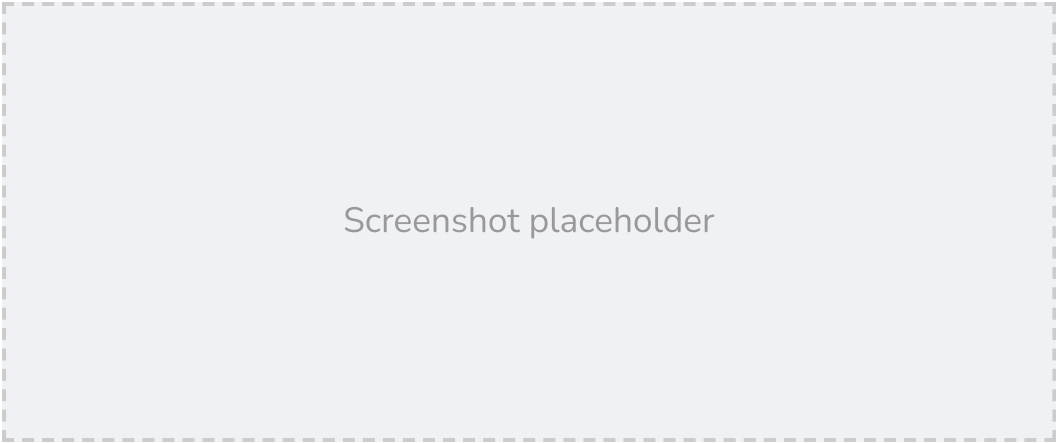
Access

Analyse

DEMO

Request

TODO: Screenshots for cohorts receiving and approving request



Placeholder for a screenshot showing cohorts receiving and approving a request.

Local data

Catalogue

Request

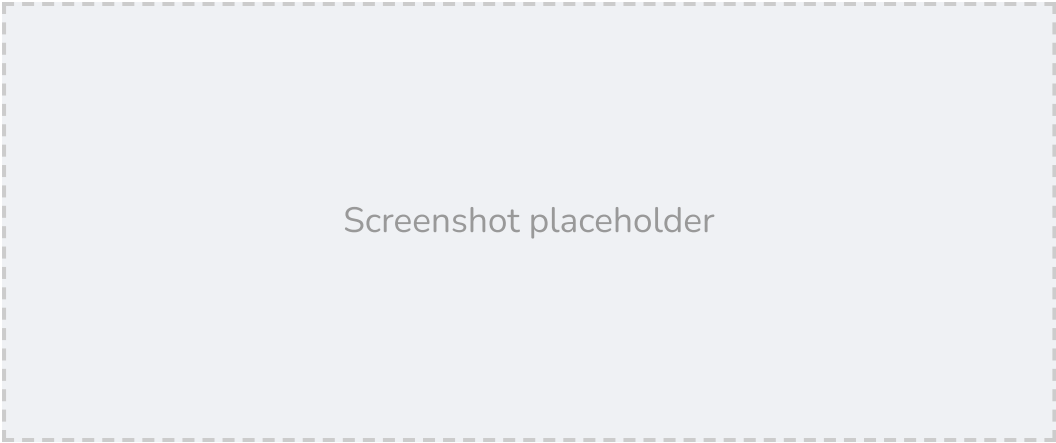
Access

Analyse

DEMO

Request

TODO(?): Process to add user automatically to Armadillo



Screenshot placeholder

Local data

Catalogue

Request

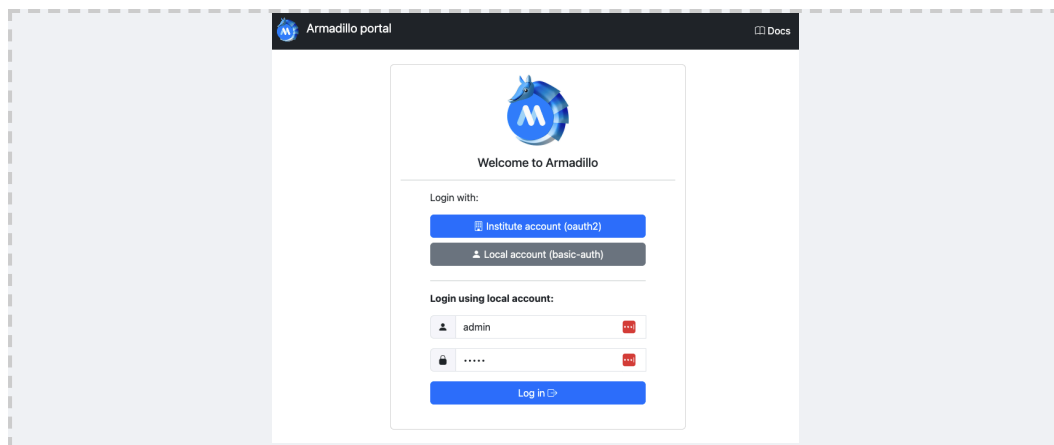
Access

Analyse

DEMO

Access

Data manager logs on to local Armadillo instance



Local data

Catalogue

Request

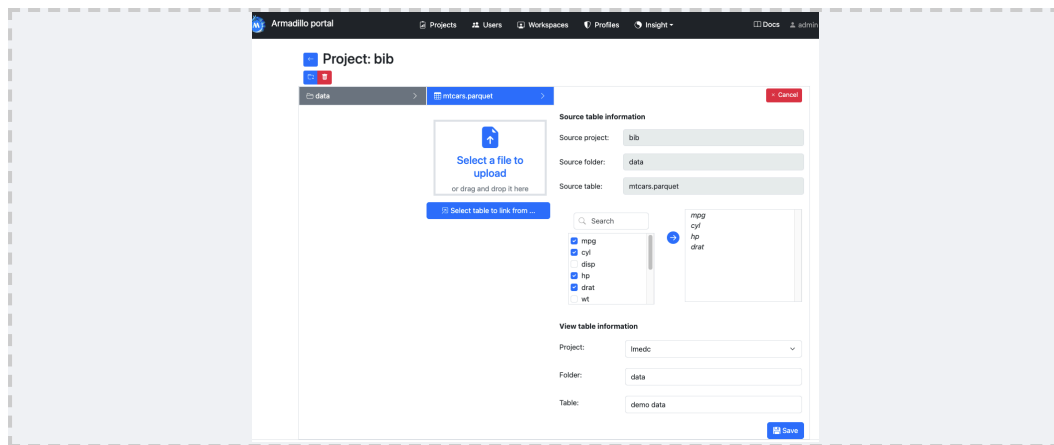
Access

Analyse

DEMO

Access

Creates view with approved variables



Local data

Catalogue

Request

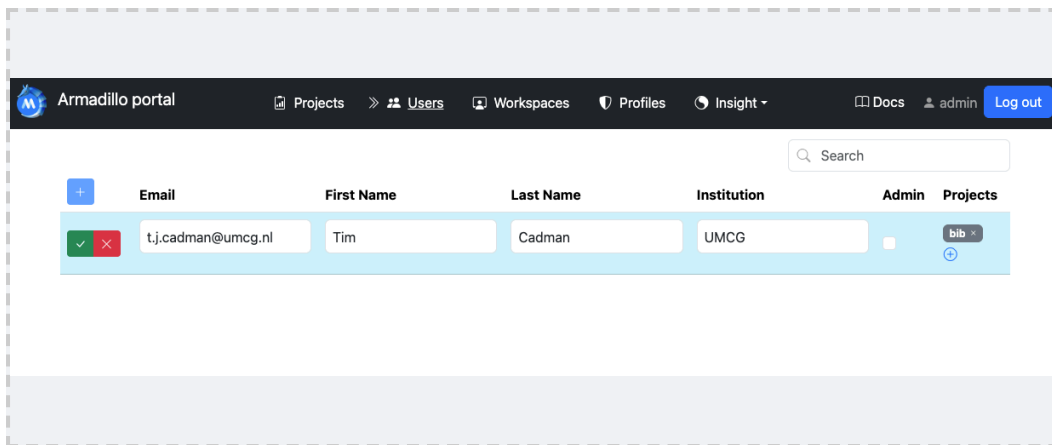
Access

Analyse

DEMO

Access

Adds user and grants them access to their project view, notifies user



The screenshot shows the 'Users' management page in the Armadillo portal. The header includes navigation links for Projects, Users, Workspaces, Profiles, and Insight, along with a search bar and a 'Log out' button. The main content area features a table with columns for Email, First Name, Last Name, Institution, Admin, and Projects. A new user entry is being added, with the email 't.j.cadman@umcg.nl', first name 'Tim', last name 'Cadman', and institution 'UMCG'. The 'Admin' column has a toggle switch, and the 'Projects' column has a dropdown menu.

	Email	First Name	Last Name	Institution	Admin	Projects
✓ ✕	t.j.cadman@umcg.nl	Tim	Cadman	UMCG	☐	bib +

Local data

Catalogue

Request

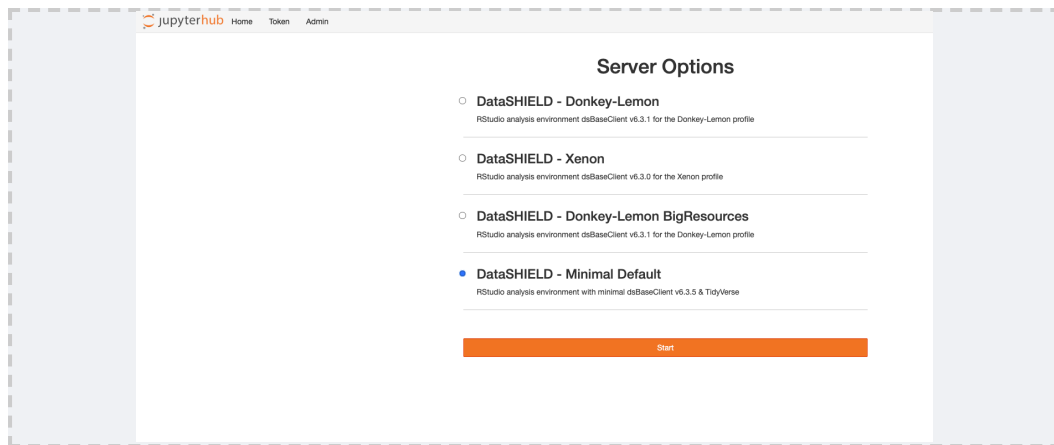
Access

Analyse

DEMO

Analyse

Researcher logs into Central Analysis Server using OIDC



Local data

Catalogue

Request

Access

Analyse

DEMO

Analyse

Researcher connects to cohorts via DataSHIELD and writes analysis scripts

```
1 #####
2 ## Project: INTERATE (Medic - Mode of Birth and Postpartum Depression)
3 ## Script purpose: Data preparation scripts
4 ## Date: January 2026
5 ## Author: Tim Cohen
6 #####
7
8 #####
9 # Maternal age
10 #####
11 ds.case_when(
12   list(
13     nonreplagebirth_m_y ~ 38 ~ "15_38",
14     nonreplagebirth_m_y ~ 31 & nonreplagebirth_m_y ~ 35 ~ "31_35",
15     nonreplagebirth_m_y ~ 35 ~ "35_47"
16   )
17 )
18 newobj = "mat_age_trp"
19
20 ds.factor("mat_age_trp", "mat_age_cat")
21
22 #####
23 # Factor status
24 #####
25 ds.filter(
26   df.name = "yearrep",
27   tidy_expr = list(age_years < 1),
28   newobj = "yearrep_8.1"
29 )
30
31 #####
32 ## Project: INTERATE (Medic - Mode of Birth and Postpartum Depression)
33 ## Script purpose: Loading derived variables
34 ## Date: January 2026
35 ## Author: Tim Cohen
36 #####
37
38 #####
39 # Drop unwanted variables
40 #####
41 nonrep_analysis_vars <- c(
42   "child_id", "head_id", "pid_label", "mat_age_cat", "mat_wig",
43   "birth_year_cat", "mar_label", "mental_health", "physical_health", "gs_cat",
44   "hospital_admission", "parity", "age_cat", "birth_weight_cat",
45   "maternal_cat", "classified_cat"
46 )
47 yearrep_analysis_vars <- c(
48   "child_id", "ada_cat", "cshab_cat"
49 )
50
51 ds.select(
52   df.name = "nonrep",
53   tidy_expr = list(!nonrep_analysis_vars),
54   newobj = "nonrep_sub"
55 )
56
57 ds.select(
58   df.name = "yearrep_8.1",
59   tidy_expr = list(!yearrep_analysis_vars),
60   newobj = "yearrep_sub"
```

Local data

Catalogue

Request

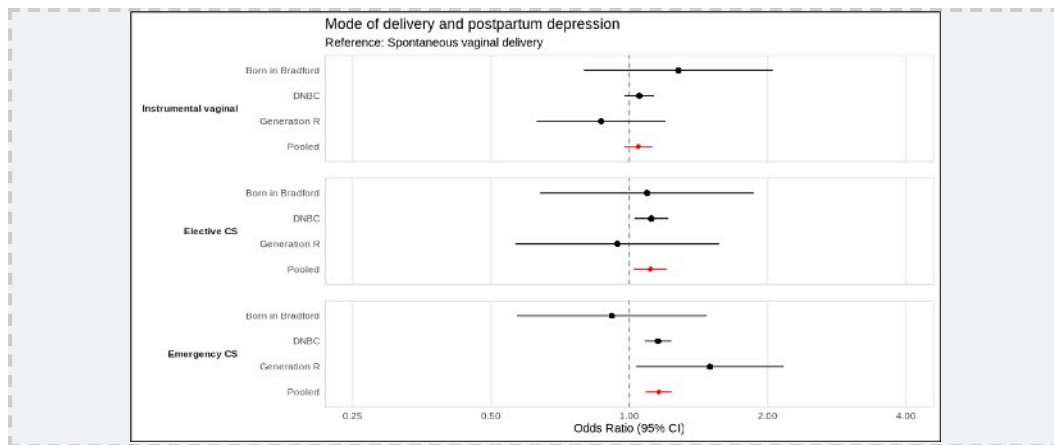
Access

Analyse

DEMO

Analyse

Preliminary results: mode of delivery and postpartum depression



Local data

Catalogue

Request

Access

Analyse